

MITRE ATLAS: Adversarial Threat Landscape for AI Systems

A Comprehensive Matrix of Adversary Tactics, Techniques, and Real-World AI Attack Case Studies

MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems) is a globally accessible, curated knowledge base of adversary tactics, techniques, and procedures (TTPs) targeting artificial intelligence and machine learning systems. Modeled after the MITRE ATT&CK framework, ATLAS provides security analysts, threat modelers, and red teams with structured taxonomies to assess real-world vulnerabilities across the end-to-end AI lifecycle.

1. MITRE ATLAS Tactical Framework Overview

ATLAS categorizes malicious actions against AI systems into 12 core tactics describing the adversary's operational objectives:

- AML.TA0000 Reconnaissance: Gathering intelligence on target AI architecture, model families, training data distributions, and API endpoints.
- AML.TA0001 Resource Development: Establishing capabilities, acquiring GPU infrastructure, fine-tuning surrogate models, or crafting adversarial payloads.
- AML.TA0002 Initial Access: Gaining entry to AI development environments, model registries, data ingestion pipelines, or inference endpoints.
- AML.TA0003 ML Attack Execution: Triggering adversarial capabilities against ML algorithms (e.g., executing direct or indirect prompt injection).
- AML.TA0004 Persistence: Maintaining access to AI assets, injecting persistent backdoors in model checkpoints, or tampering with agent memory.
- AML.TA0005 Defense Evasion: Crafting inputs that evade ML-based intrusion detection, input guardrails, or safety content filters.
- AML.TA0006 Discovery: Exploring model boundaries, extracting system prompts, inspecting vector database schemas, and mapping agent tools.
- AML.TA0007 Lateral Movement: Exploiting autonomous agent permissions to pivot from an LLM interface into enterprise databases and cloud IAM.
- AML.TA0008 Collection: Extracting private training data, intellectual property, confidential RAG documents, or sensitive user completions.
- AML.TA0009 ML Model Access & Exfiltration: Stealing model weights, replicating functionality via black-box distillation, or extracting vector embeddings.
- AML.TA0010 Impact: Degrading model availability via denial-of-service, manipulating automated trading decisions, or causing reputational damage.

2. Key ATLAS Techniques for Large Language Models and Generative AI

ATLAS defines specific technical procedures commonly executed against GenAI and LLM architectures:

- AML.T0051 LLM Prompt Injection: Crafting adversarial strings that trick an LLM into ignoring system safety guardrails. In indirect prompt injection (AML.T0051.001), untrusted data retrieved from external sources contains malicious instructions that seize control of the execution flow.

- AML.T0054 LLM System Prompt Extraction: Interrogating an LLM through recursive querying and framing techniques to reveal proprietary system prompts, hidden rules, and confidential credentials.
- AML.T0040 ML Model Extraction: Querying a proprietary commercial model systematically to train a local surrogate model that replicates the original model's decision boundaries without paying licensing fees or respecting access restrictions.
- AML.T0043 Insecure Tool Execution in Autonomous Agents: Exploiting excessive permissions granted to LLM function calling and agent tools to execute arbitrary system commands, delete cloud storage, or initiate unauthorized transactions.
- AML.T0048 LLM Hallucination Exploitation: Inducing the LLM to generate fictitious package names (package hallucination attack), leading developers to install rogue packages registered by adversaries on PyPI or npm.

3. Threat Modeling AI Systems using MITRE ATLAS

Threat modeling for artificial intelligence applications requires extending traditional stride/threat trees to account for unique AI failure modes:

- Step 1 - Asset Identification: Enumerate critical AI assets including training datasets, pre-trained weights, fine-tuning adapters (LoRA), prompt templates, vector embeddings, and API tokens.
- Step 2 - Adversary Capability Assessment: Determine whether the adversary operates under White-Box assumptions (full access to model weights and architecture), Black-Box assumptions (API query access only), or Gray-Box assumptions (partial access to embeddings or tokenizer).
- Step 3 - Attack Surface Mapping: Map each component of the AI architecture: (1) Data ingestion, (2) Model training and fine-tuning, (3) RAG vector retrieval pipeline, (4) LLM inference runtime, and (5) Downstream agent tools and integrations.
- Step 4 - Threat Vector Analysis: Overlay ATLAS techniques against mapped components to identify high-likelihood attack paths (e.g., indirect prompt injection via untrusted PDF parsing).
- Step 5 - Mitigation Specification: Select and verify defensive controls mapped directly to ATLAS mitigations (e.g., AML.M1004 Privileged Process Integrity, AML.M1005 Model Hardening).

4. Real-World Case Studies in MITRE ATLAS

Documented incidents demonstrating how adversaries attack real-world machine learning systems:

- Case Study 1 - Indirect Prompt Injection in Bing Chat / Copilot: Researchers demonstrated that hidden text on a public webpage could instruct the Bing search AI to steal personal user data and trick users into visiting phishing domains.
- Case Study 2 - Proofpoint Email Protection ML Evasion: Attackers repeatedly queried an enterprise email filtering model to compute the classification decision boundary, subsequently discovering formatting perturbations that allowed malicious emails to bypass the spam filter entirely.
- Case Study 3 - Poisoned HuggingFace Model Weights: Security researchers discovered hundreds of malicious machine learning repositories containing PyTorch pickle files that executed reverse shells and remote code upon being loaded into memory.